Design Milestones

Initial Research Proces

1.General Research

I started my project by researching different types of hot beverage devices. After I decided that I want to focus on a coffee preparation device I asked AI to list different types of coffee machines and types of the coffees.

The list that AI made:

- Drip Coffee Makers
- Espresso Machines (Manual, Semi-Automatic, and Super-Automatic)
- French Press
- Moka Pot
- Pour-Over (V60/Chemex)
- AeroPress
- Siphon (Vacuum) Brewer
- Percolator
- Turkish Coffee Pot (Cezve/Ibrik)

Reflection: The list helped me to see on which types of machines I can focus. I accepted the drip coffee makers and decided to continue on them.

Main Focus

2.Interwiew- Context Change

In the beginning of my concept selection I wanted to concentrate on the shared student housing. After a while I decided to shift my focus from dormitory-friendly usage to ferry-friendly usage, as I find it more interesting and professionally improving. To support this transition, I conducted an interview with a friend who works as a crew member on a ferry.

Two questions that I asked shaped my design decisions:

5. What feels unsafe or messy?
Everything that runs on a plug feels generally unsafe because the current on the ship is so strong; there is a constant fear of being electrocuted.

6. If you could change one thing about the process, what would it be?
I wish the devices were much more stable; since the ferry is constantly in motion, we have to wedge all the machines tightly against each other. Otherwise, they either tip over or the liquid spills out.

3. Research With AI

After the interview I also asked AI what risks and electrical standards there are in the ships and what can I focus on while designing. AI helped me understand the specific electrical and safety standards for marine environments(IP ratings and voltage limits).Standard home appliances would not work because of the constant vibration and ship motion.

I accepted the suggestion to use bolt-down mountings because they are essential for durability and safety at sea. I rejected the idea of a complex 3-phase power system because we need devices to be more practical.

4. Deciding the Context

The interview and research that I conducted with AI helped me to shape my user and context.

User: Crew members who are working on ferries.
Context: weekdays, tiny kitchen that is full of other devices, on the see

Observations:
-Space: Extremely limited space requiring a compact footprint.
-Stability: High-motion environment causing spills and tipping.
-Safety: High risk of shocks and surges because of the electricity.
-Utility: Rapid, low-effort stimulant delivery for alertness.
-Users drink coffee to stay awake and function.

Insights:
In a moving ship and a busy work schedule, a tired crew member needs a very fast and safe way to get caffeine. However, the fear of electric shocks, the risk of hot spills from waves, and the lack of space make it stressful to prepare a simple drink while trying to stay alert and focused on the job.

Opportunity Areas:
-Lower spillings by making the device more stable.
-Improve the ergonomy of the device ona right handed way. (Crews work with only their right hands on ships because it is farther from the heart, as high electrical currents on vessels can cause serious harm.)
-Smaller design in order to fit the compact kitchen without taking much of a space.

“I will design for ship crews working in small and unstable sea environments by focusing on the risk of electrical shocks and spilling liquids because it causes dangerous accidents and creates stress.”
-I rejected portable or standard machines because they are too unstable to use during waves and can easily cause electrical fires

Initial Design Process

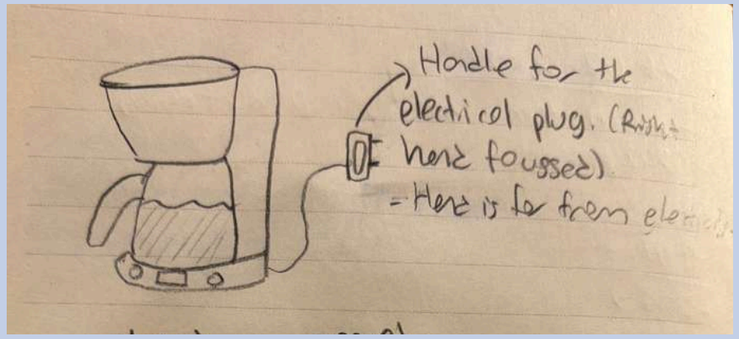
I started with making sketches for my device focusing on three main tensions: electricity, spilling and stability. I uploaded my basic sketches to AI (Gemini) and asked it to visualize them.

Prompts and visual attachments that I gave to AI:

Visual outputs that AI gave me

Promt: Can you visualize this filter coffee machine? It looks like a regular filter coffee machine but it actually has a plug like a handle in order to prevent electricity based accidents.

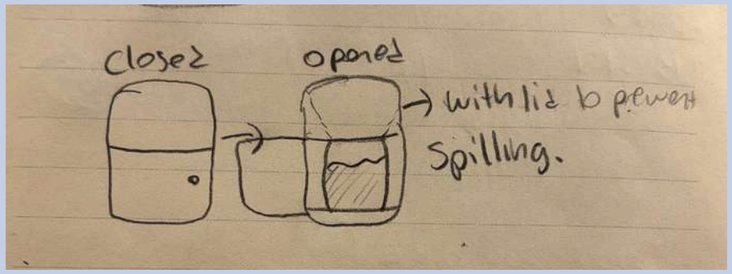
Attachment



Electrical Safety

Promt: Can you visualize this sketch it doesnt looks like a regular filter machine it has lid in the place that the coffee fills in order to prevent slipping when it is closed it looks like a box but when it opens you can see the coffee and take and spill it

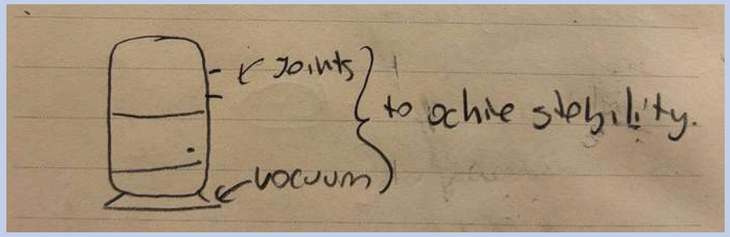
Attachment



Spill Safety

Promt: Can you visualize this? It is a filter coffee machine but it has vacuums under it in order for it to be stable in ships and it also has joints as a second stabilization. The joints join the walls but this coffee machine has a door like the other one so when it is closed it looks like a box as well.

Attachment



Spill and Stability Safety

Reflection: I accepted the images but I realized that I want all of the concepts in one single design in order to minimize the risks in the ships.

Design Process

Promt: Can you visualize the filter coffee machine in this sketch? This filter coffee machine will be made for the crew on ships. It will have a lid so that the contents won't spill. It will have vacuum seals at the bottom so that it won't tip over from the counter. And the part that plugs in will be like a handle so that the hand will be further from the electrical current.

Attachments

1-Spill and Stability Safety

+

2-Electrical Safety

=

AI visual output

3-Spill Safety

Instead of rejecting the concepts I decided to integrate them in one single device. Because I think the dangerous and difficult marine environment and circumstances need a more integrated system that has all three concepts.

Overall, this improved innovation removes the dangers of a maritime kitchen through an engineering design that provides mechanical stability, as well as protection for the physical well being of the crew.

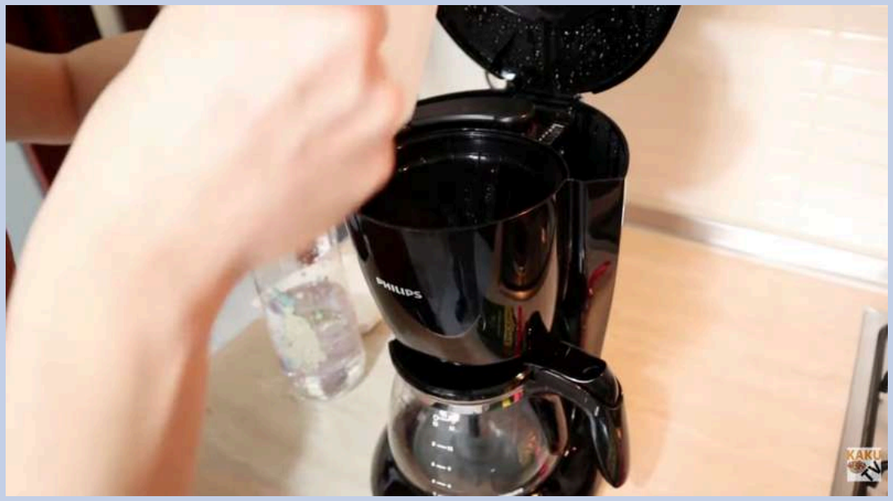
The novel use of a vacuum-suction base, a spill-proof lid and a right-handed optimized safety closure collectively provide continuity of service during oceanic vibrations while protecting the crew of 6 from electrical and thermal accidents.

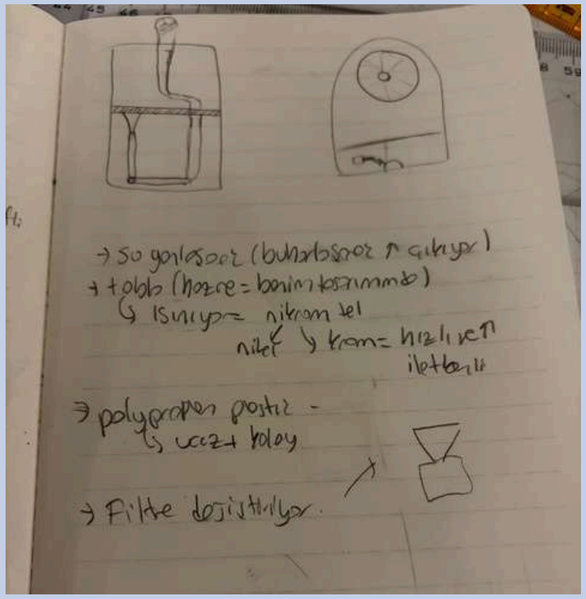
Change: After I saw my design visually I realized that the door system takes much more space than I thought and I decided to make a sliding door in order to maintain space.

Technical Research

I wanted to make a prototype for my design in order to understand the way that it actually works. To make a 3D prototype I had to understand the mechanical side of the machine so I watched 2 videos and took some notes from them. I added the mechanical properties to my design.

Screenshots of the main scenes of the video that I learned how the machine works from outside. .





Screenshots of the main scenes of the video that I learned how the machine works from inside. .

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Prototyping

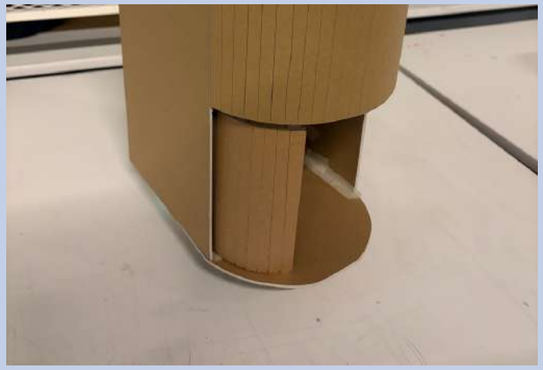
I made a prototype to better understand the 3D version of my device. I realized that the measurements that were in my head are too much and I need to make it smaller. Then I uploaded the pictures of my prototype that I took from different angles and wrote a prompt asking it to visualize my design. I also changed the

Attachment- Prototype

Outside

Inside

Filter



Spill Safety Sliding door

vacooms

Electrical Safety Handle Plug

Promt:

I am a second-year student at Özyeğin University and I am designing a filter coffee machine as part of the IDE 202 program. This filter coffee machine will be used by people working on ferries or ships, so to prevent it from falling, it will have three vacuum seals at the bottom. The bottom part will be like a sliding door so that the coffee doesn't spill out. I also envisioned using a washable filter. Could you visualize the prototype I sent based on these specifications? The machine's height from the ground is 29 cm (excluding the vacuum seals), 18 cm wide, and 18 cm long.

AI visualizations

Reflection:

In the prototype I had 3 vacuums but AI visualized 4 and I accepted them because I think it is more steady and accurate for solving the stability problem. AI placed the water tank on the side of the machine by making it transparent. I liked the idea that we see the water so I decided to keep that as well. I didn't like the way that plug looks. It looks unsafe and unergonomic so I decided to make a regular plug.

Final Design Decision

After several design stages, I decided to use a faucet system for dispensing coffee instead of a traditional carafe. This approach minimizes the risk of spills and reduces the physical effort required to pour coffee. The machine operates by drawing water from the reservoir through a small opening into internal tubing. At the base, a nichrome wire serves a dual purpose: it keeps the brewed coffee warm and heats the water tubes simultaneously. As the water boils and turns into steam, it rises through the secondary pipe, condenses at the showerhead, and drips evenly over the coffee grounds. Once filtered, the coffee collects in a durable metal reservoir.

To address hygiene concerns, the faucet is detachable and includes a removable dip tube that ensures no coffee is wasted; both are easy to wash, keeping the entire liquid path sanitary. For stability, the base features vacuum suctions with integrated air-release tabs, allowing the machine to stay firmly in place during rough seas while remaining easy to relocate. Combined with a reusable filter for sustainability, this design is perfectly tailored for both ferry crews and long-haul maritime staff, providing a safe and low-maintenance solution at sea.

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